StudDFT keywords
(short manual, v0.82.1)

Keyword	Options	Synonyms	Designation
Quantum chemical theories			
%method			Required. Electronic-structure method (case-insensitive).
	HF		Hartree-Fock
	MP2		HF reference + MP2 correlation (post-SCF)
	SVWN	SVWN5, LDA, LSDA	Slater + VWN-V LDA (NWChem / PySCF convention)
	SVWN3		Slater + VWN-III RPA (Gaussian 09 convention)
	BLYP		Becke88 + LYP GGA
	BP86	BP	Becke88 + Perdew86 GGA
	BPW91		Becke88 + Perdew-Wang 91 GGA
	PBE		Perdew-Burke-Ernzerhof GGA
	PBE0	PBE1PBE	PBE with 25 % HF exchange
	B3LYP	B3LYP5	B3LYP with VWN-V correlation (ORCA / NWChem)
	B3LYP3		B3LYP with VWN-III correlation (Gaussian convention)
			Note: open-shell calculations automatically use U-prefixed variants (UHF, UMP2, UB3LYP, ...).

<table>
<tr><td colspan="4">Job type</td></tr>
<tr><td>%job</td><td></td><td></td><td>Task to run. Default: energy.</td></tr>
<tr><td></td><td>energy</td><td>sp, single</td><td>Single-point energy (default)</td></tr>
<tr><td></td><td>gradient</td><td>grad, force</td><td>Analytical gradient at fixed geometry</td></tr>
<tr><td></td><td>opt</td><td>optimize</td><td>Geometry optimization (internals RFO or Cartesian BFGS, see %opt_coord)</td></tr>
<tr><td></td><td>freq</td><td>frequency, hessian</td><td>Hessian + harmonic vibrational frequencies</td></tr>
<tr><td></td><td>opt+freq</td><td>optfreq</td><td>Opt followed by freq at the optimized geometry</td></tr>
<tr><td></td><td>bsse</td><td>counterpoise, cp</td><td>Boys-Bernardi counterpoise (requires fragment IDs on every geometry line)</td></tr>
<tr><td></td><td>scan</td><td>pescan, rigid_scan</td><td>Rigid PES scan; requires %gz block and %scan ... *end block</td></tr>
<tr><td></td><td>scanopt</td><td>relaxed_scan, scan+opt</td><td>Relaxed PES scan: at every grid point a constrained geometry optimization is performed</td></tr>
<tr><td></td><td>ts</td><td>opt_ts, optts</td><td>Transition-state search via P-RFO (HF / DFT only)</td></tr>
<tr><td></td><td>ts+freq</td><td>tsfreq, opt_ts+freq, opttsfreq</td><td>TS search followed by freq pass at located TS</td></tr>
<tr><td colspan="4">Molecule builder</td></tr>
<tr><td>%charge</td><td><signed integer></td><td></td><td>Total molecular charge. Default: 0.</td></tr>
<tr><td>%mult</td><td><positive integer></td><td>%multiplicity</td><td>Spin multiplicity 2S+1 (1 = singlet, 2 = doublet, ...). Default: 1.</td></tr>
<tr><td>%spin</td><td>restricted | unrestricted</td><td>RHF, RKS, R | UHF, UKS, U</td><td>SCF spin treatment. If not set, determined automatically from %mult.</td></tr>
<tr><td>%geo</td><td>[angstrom | bohr]</td><td>%geometry, *geometry</td><td>Start of Cartesian geometry block. Terminated by *end or</td></tr>
</table>

			empty line. Atom token: <Z> | <Element> | <Element><index>.
%gz			Start of Z-matrix geometry block (Angstroems, degrees). Terminated by empty line. Required for %job scan / scanopt.
Basis set			
%basis	<basis set name>		Required. Basis set name (case-insensitive). Built-in: sto-3g, 3-21g, 6-31g, 6-31g(d), 6-31g(d,p), 6-31+g(d), 6-31++g(d,p), 6-311g(d), 6-311g(d,p), 6-311g(2d,2p), 6-311g(3df,2pd), 6-311++g(2d,2p), 6-311++g(3df,2pd), cc-pVTZ, cc-pVQZ, aug-cc-pVDZ, aug-cc-pVTZ, aug-cc-pVQZ, LANL2DZ. MIXED: per-atom assignment via <selector> <basis_name> lines until next %-keyword or empty line.
%spherical	(flag)		Use spherical-harmonic d/f-shells (5d/7f); default is Cartesian (6d/10f).
ECP			
%ecp	<ecp name>		Effective core potential. Built-in: LANL2DZ. MIXED: per-atom assignment via <selector> <ecp_name> lines until next %-keyword or empty line.
%ecp_method	[numerical] | analytical	numerical: quadrature, num, 0 | analytical: md, mcmurchie, 1	ECP integral evaluation method. Default: numerical (Mura-Knowles radial quadrature). Analytical = McMurchie-Davidson; substantially overhauled in v0.77.0-0.77.1 (now matches g09 to ~1 microHa on AlH3-

			class molecules). For very tight basis primitives (alpha > ~5000) prefer numerical.
Electronic integral calculations			
%boys_cook	(flag)	%boys_precise	Use Cook recurrence for Boys function instead of fast tabulated representation. Default: off (table method).
SCF procedure			
%scf_converger	[DIIS] | NONE	%converger	Convergence accelerator. Default: DIIS. Internally tests whether value starts with "DIIS"; anything else turns DIIS off.
%scf_maxiter	<positive integer>		Max number of SCF iterations. Default: 150.
%scf_etol	<positive real>		Energy convergence threshold (Hartree). Default: 1e-8.
%scf_ptol	<positive real>		Density-matrix convergence threshold. Default: 1e-6.
%scf_diis_etol	<positive real>	%scf_gtol	DIIS error-norm convergence threshold (v0.76.0). All three (%scf_etol, %scf_ptol, this one) must be satisfied simultaneously. Default: 1e-5.
%scf_min_iter	<positive integer>		Minimum SCF iterations before convergence can be declared (v0.76.0). Default: 5.
%scf_n_converged	<positive integer>	%scf_nconv	Consecutive iterations all three convergence criteria must be satisfied (v0.76.0). Default: 2.
%scf_diis_cond_max	<real > 1>	%scf_diis_condmax	Upper bound on DIIS B-matrix condition number; oldest vector is pruned when exceeded (v0.76.0). Default: 1e10.

<table>
<tr><td>%damping</td><td><real in [0,1]></td><td></td><td>Density-matrix damping factor. DIIS overrides damping after a few iterations. Default: 0.0.</td></tr>
<tr><td>%level_shift</td><td>AUTO | OFF | <non-negative real></td><td></td><td>Virtual-orbital level shift (Hartree). Useful for problematic open-shell / heavy-element cases. Default: AUTO (turning on shift 0.2 if SCF is not converged in 100 steps)</td></tr>
<tr><td>%initial_guess</td><td>AUTO | SAD | CORE</td><td>%guess, %scf_guess, HCORE, CORE_H</td><td>Initial-guess strategy (v0.76.0). Default: RHF/RKS - CORE, UHF|UKS - AUTO (SAD if a fit basis is loadable, CORE otherwise).</td></tr>
<tr><td>%uks_occupation</td><td>AUTO | STRICT | MOM | MIX</td><td>%uhf_occupation, %occupation, AUFBAU, MAXOV, MAX_OVERLAP, REMIX</td><td>UHF/UKS orbital-occupation strategy (v0.76.0). Default: AUTO. No effect for restricted.</td></tr>
<tr><td>%eri_memory_limit_mb</td><td><positive integer, MB></td><td>%eri_mem_limit_mb, %eri_memory_limit, %eri_mem_limit</td><td>ERI-buffer memory cap (v0.76.3); falls back to direct JK when exceeded. Default: 4096.</td></tr>
<tr><td colspan="4">DFT options</td></tr>
<tr><td>%grid</td><td></td><td>%grid_quality</td><td>DFT integration-grid quality. Default: 3 (fine, 75 radial). No effect for HF or MP2.</td></tr>
<tr><td></td><td>1</td><td></td><td>Coarse (25 radial)</td></tr>
<tr><td></td><td>2</td><td></td><td>Medium (50 radial)</td></tr>
<tr><td></td><td>3</td><td></td><td>Fine (75 radial; default)</td></tr>
<tr><td></td><td>4</td><td></td><td>Ultrafine (100 radial)</td></tr>
<tr><td></td><td>5</td><td></td><td>Extreme (150 radial)</td></tr>
<tr><td></td><td>fine</td><td>gfine</td><td>Preset 6: Gaussian Fine (75 radial, 302 Lebedev, Euler-Maclaurin + size-adjustment)</td></tr>
<tr><td></td><td>ultrafine</td><td>gultrafine</td><td>Preset 7: Gaussian UltraFine (99 radial, 590 Lebedev)</td></tr>
</table>

	tm	treutler	Preset 8: Treutler-Ahlrichs (75,302) uniform 302-point angular, no pruning (replaces former pyscf alias in v0.77+)
	g09	g09ref	Preset 9: G09 reference (75,302) uniform 302-point angular, Euler-Maclaurin + size-adjustment
%xc_deriv	[analytical] | numerical	analytical: anal, exact | numerical: num, fd	Analytical or finite-difference XC functional derivatives during SCF and DFT gradient / Hessian. Default: analytical.
Dispersion			
%dispersion	NONE | D3	%disp, D3BJ, D3-BJ	Empirical dispersion correction. Default: NONE. D3(BJ) params: HF, BLYP, BP86, BP, BPW91, B3LYP*, PBE, PBE0.
MP2 options			
%frozen_core	-1 | 0 | <positive integer>	%nfrozen	Frozen-core control. -1 (default) = auto-detect noble-gas core of every non-ECP atom; 0 = correlate all; N = freeze lowest N occupied orbitals (per spin for UMP2).
%pump2	[yes] | no	on, true, 1, T, <bare> | off, false, 0, F	Spin-projected open-shell MP2 (Schlegel-1986 single-annihilation). Default: yes -- PUMP2 single-point diagnostic always printed when UMP2 is active. With %pump2 yes, FD-PUMP2 optimization is also enabled (gradient w.r.t. PUMP2 energy). With %pump2 no, PUMP2 is fully suppressed.

<table>
<tr><td colspan="4">Geometry optimization</td></tr>
<tr><td>%geom_maxiter</td><td><positive integer></td><td>%opt_maxiter, %opt_max_iter, %geom_max_iter</td><td>Max optimization steps. Default: 50.</td></tr>
<tr><td>%grad_tol</td><td><positive real></td><td></td><td>Maximum-component gradient convergence threshold (Ha/Bohr). Default: 4.5e-4.</td></tr>
<tr><td>%grad_method</td><td>[analytical] | semianalytical</td><td>analytical: full, 1 | semianalytical: semi, fd, 0</td><td>Gradient method. PUMP2 gradient is always numerical (h = 1e-3 Bohr).</td></tr>
<tr><td>%opt_coord</td><td>[internal] | cart</td><td>%opt_coordinates, %opt_coords | internal: internals, redundant, rfo, int | cart: cartesian, bfgs, cart_bfgs, cartesian_bfgs, xyz</td><td>Optimizer coordinate system (v0.75.0). Default: internal (redundant-internals RFO). Cartesian path does NOT support constraints or PCM.</td></tr>
<tr><td colspan="4">Geometry constraints (partial-constraint optimization)</td></tr>
<tr><td>%constrain</td><td></td><td></td><td>Start a constraint block (v0.70+). One declaration per line, terminated by *end or empty line. Compatible with %job opt / opt+freq, internal-coordinate optimizer only.</td></tr>
<tr><td></td><td>B <i> <j></td><td></td><td>Freeze bond between atoms i and j</td></tr>
<tr><td></td><td>A <i> <j> <k></td><td></td><td>Freeze angle i-j-k (vertex at j)</td></tr>
<tr><td></td><td>D <i> <j> <k> <l></td><td></td><td>Freeze dihedral i-j-k-l</td></tr>
<tr><td colspan="4">PES scan (rigid / relaxed)</td></tr>
<tr><td>%scan</td><td></td><td></td><td>Start a sweep-variable block (v0.69+). Required by %job scan and %job scanopt. One sweep variable per line, terminated by *end or empty line. Requires a %gz Z-matrix geometry.</td></tr>
<tr><td></td><td><kind> <row> [from] <val_from> [to] <val_to> [step] <val_step></td><td></td><td><kind> = R (bond) | A (angle) | D (dihedral); <row> = 1-based Z-matrix row index. Keywords from/to/step are optional.</td></tr>
</table>

Hessian / Frequencies			
%hess_method	[analytical] | numerical	analytical: full, 1 | numerical: num, fd, 0	Hessian computation method. Default: analytical (HF, DFT, ECP). Numerical = finite difference on analytical gradient (h = 5e-3 Bohr).
TS options (transition-state search)			
%read_hess	<filename>	%read_hessian	.hess file to load before TS search (v0.72.0). Reader verifies geometry, method, basis, ECP, charge, multiplicity, spin, format version. If omitted, the TS driver computes the Hessian itself at the input geometry.
%ts_mode	<positive integer>	%tsmode	1-based mode index to follow on the first P-RFO step (v0.72.0). After step 1, optimizer switches to maximum-overlap mode tracking. Default: 1 (lowest mode).
PCM (implicit solvent, v0.73.0+)			
%pcm	none | cpcm | iefpcm | cosmo	cpcm: c-pcm, c_pcm | iefpcm: ief-pcm, ief_pcm, pcm | none: off, no, false, 0, F	Continuum solvation model. Default: none. %solvent or %eps auto-activate CPCM if no %pcm given.
%solvent	<name>		Named solvent (resolved to tabulated permittivity; overrideable by %eps). Recognized: water (h2o), methanol (meoh), ethanol (etoh), acetone, acetonitrile (mecn), dmso, dmf, thf, chcl3 (chloroform), ccl4, dichloromethane (dcm, ch2cl2), toluene, benzene, cyclohexane, hexane (n-hexane), diethylether (et2o, ether), ethyl_acetate (etoac), ammonia (nh3), vacuum (none).

%eps	<real >= 1.0>	%epsilon, %pcm_eps	Direct override of dielectric permittivity. Vacuum = 1.0.
%pcm_lebedev	<positive integer>		Per-atom Lebedev quadrature order on cavity surface. Default: 110. Allowed: 6, 14, 26, 38, 50, 74, 86, 110, 146, 170, 194, 230, 266, 302, 350, 434, 590.
%pcm_scaling	<positive real>		Scaling factor on Bondi vdW radii for atomic-sphere cavity. Default: 1.20 (standard COSMO/CPCM).
%pcm_switch	<positive real, Bohr>		Width of cubic-taper switching function on cavity-tessera occupancy. Default: 0.20 Bohr (typical 0.1 - 0.4).
%pcm_grad_method	[analytical] | numerical	%pcm_gradient_method | analytical: ana, an, 1, true, yes, on | numerical: num, fd, 0, false, no, off	PCM gradient method (v0.74.x). Default: analytical (fast). Numerical = central-difference FD, ~10-30x slower; agrees with analytical to ~1e-5 a.u./bohr.
Output control			
%print_mo	(flag)		Print converged MO coefficient matrix at end of SCF.
%print_time	(flag)	%print_timing	Print detailed timing breakdown per timer counter at end of run.
%print_grad	(flag)	%print_gradient	Print per-atom Cartesian gradient at every optimization step (format: <Element><index> gX gY gZ).
%export_mo	(flag)		Save converged MO coefficient matrix in Molden format.to the external file <base>_mo.molden

Comments			
# or !			Comment markers at start of line. Whole line is ignored. Inline comments are NOT supported -- everything after the keyword on the same line is parsed as the value.